

is that for most of them, the exact values are known only for some values of the argument, while for all other points, their values are calculated numerically or by using other approximations. Almost all special functions can be written in one of two ways (which can be proven), and based on that, they can be classified into two groups:

- functions in the form of integrals,
- functions in the form of infinite convergent (usually power) series.

Functions in the form of integrals include beta and gamma functions, which, among other applications, are important in analysing other special functions. The beta and gamma functions are also known as Euler's integral of the first and second kind, respectively.

Given their widespread application in engineering, there is a need for software capable of efficiently approximating the values of these functions – for example, an intuitive web application that will provide the approximation in a minimal number of steps, generate a graphical representation of the function, and provide basic function information. This is the core idea behind the web application at hand.

2 Overview of Implemented Functions

This part of the paper gives a brief overview of the special mathematical functions of which approximations are implemented in the subject application.

2.1 The Bessel function of the first kind

Let n be an integer non-negative constant. Then the function

$$J_n(x) = \sum_{m=0}^{\infty} (-1)^m \frac{\left(\frac{x}{2}\right)^{2m+n}}{(m+n)! \cdot m!} \quad (1)$$

is called a Bessel function of the first kind, order α , with argument x , $x \in \mathbb{R}$.

The Bessel functions of the first kind are defined as the solutions to the Bessel differential equation, which is a homogeneous linear differential equation of the second order [6]:

$$x^2 y'' + xy' + (x^2 - \alpha^2)y = 0. \quad (2)$$

Generating function of Bessel function of the first kind is:

$$g(x, t) = \sum_{n=-\infty}^{\infty} J_n(x) t^n = e^{\frac{x}{2} \left(t - \frac{1}{t} \right)}. \quad (3)$$

Bessel function of the first kind, as well as most special functions, can be given using the contour integral as [10]:

$$J_n(z) = \frac{1}{2\pi i} \oint e^{\frac{z}{2}\left(t - \frac{1}{t}\right)} t^{-n-1} dt. \quad (4)$$

2.2 The Gamma function

The gamma function is one of the most commonly used special functions. It is used in number theory, probability, integral calculus, and especially in solving problems in physics. It frequently appears in the theory of other special functions, giving it particular significance. It stands out from the other functions discussed in this paper due to the simplicity of its recurrent relations, a crucial feature that arguably defines its important role in the theory of special functions.

The gamma function, on the domain of positive real numbers, is commonly defined by the Euler integral:

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt. \quad (5)$$

The gamma function is related to the factorial function for argument' integer values via the formula [3]:

$$\Gamma(n) = (n-1)!. \quad (6)$$

2.3 The Beta function

The beta function is also known as the Euler integral of the first kind. It finds its application in various fields, most often in mathematical analysis, probability theory, statistics, quantum physics and, of course, engineering.

Beta function is two variable one, and defined using a definite integral:

$$B(p+1, q+1) = \int_0^1 t^p (1-t)^q dt, \quad p, q \in \mathbb{R}, \quad p, q > 0. \quad (7)$$

This function can be expressed using the gamma function as (8) [10]:

$$B(p+1, q+1) = \frac{\Gamma(p) \cdot \Gamma(q)}{\Gamma(p+q)}. \quad (8)$$

2.4 Orthogonal polynomials

Several types of classical orthogonal polynomials exist, some of which were already mentioned in the introduction, including Laguerre, Hermite, and Jacobi polynomials, as well as Legendre and Chebyshev polynomials (both of the first and second kinds). All the aforementioned are classified as classical orthogonal polynomials, which were derived from solving linear differential equations of the second order (of the Fuchs' type) [4].

Legendre polynomials are orthogonal polynomials with a weight function $w(x) = 1$ over the interval $I = (-1, 1)$, series expansion of which is [8]:

$$P_n(x) = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k}^2 (x-1)^{n-k} (x+1)^k. \quad (9)$$

Legendre differential equation is a linear differential equation of second order [2]:

$$(1-x^2)y'' - 2xy' + k(k+1)y = 0. \quad (10)$$

Laguerre polynomials are orthogonal polynomials with a weight function $w(x) = e^{-x}$, defined over the domain of real numbers, expressed by Rodrigues's formula as:

$$L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n} (x^n e^{-x}) = \frac{1}{n!} \left(\frac{d}{dx} - 1 \right)^n x^n, \quad (11)$$

and they are solutions to the Laguerre differential equation, which is also a linear differential equation of second order [1]:

$$xy'' + (1-x)y' + \lambda y = 0. \quad (12)$$

We distinguish between two types of Hermite polynomials – physicist's and probabilist's. Hermite physicist's polynomial is the polynomial:

$$H_n(x) = n! \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^k \frac{(2x)^{n-2k}}{k!(n-2k)!}, \quad (13)$$

orthogonal with respect to the weight function $w(x) = e^{-x^2}$ over the interval $(-\infty, \infty)$. Hermite probabilist's polynomial are given by:

$$H_{e_n}(x) = 2^{-\frac{n}{2}} H_n\left(\frac{x}{\sqrt{2}}\right). \quad (14)$$

Hermite physicist's and probabilist's polynomials are solutions to the next two second order linear equations, respectively [10]:

$$\begin{aligned} y'' - xy' + ny &= 0, \\ y'' - 2xy' + ny &= 0. \end{aligned} \quad (15)$$

Chebyshev polynomials are sequences of orthogonal polynomials defined by sine and cosine functions³. We differentiate between Chebyshev polynomials of the first kind and Chebyshev polynomials of the second kind. Chebyshev

³In general, the sine and cosine functions are also considered special functions derived from this equation. However, due to their extensive and frequent use in mathematics, they are classified as elementary functions.

polynomials of the first kind, orthogonal with respect to the weight function $w(x) = (1-x^2)^{-1/2}$ over the interval $I = [-1, 1]$, are given by the expression:

$$T_n(x) = \cos(n \arccos x), \quad |x| \leq 1, n \in \mathbb{N}_0. \quad (16)$$

Chebyshev polynomials of the second kind, with weight function $w(x) = (1-x^2)^{1/2}$ over the same interval, are given as:

$$U_n(x) = \frac{\sin((n+1)x)}{\sin x}. \quad (17)$$

The Chebyshev differential equations, of which solutions are Chebyshev polynomials of the first and second kind, are respectively [10]:

$$\begin{aligned} (1-x^2)y'' - xy' + n^2y &= 0, \\ (1-x^2)y'' - 3xy' + n(n+2)y &= 0. \end{aligned} \quad (18)$$

Finally, Jacobi polynomials, with real parameters $\alpha > -1$, $\beta > -1$, are sequences of polynomials which have the following form:

$$P_n^{(\alpha, \beta)}(x) = \frac{(-1)^n}{2^n n!} (1-x)^{-\alpha} (1+x)^{-\beta} \frac{d^n}{dx^n} [(1-x)^{\alpha+n} (1+x)^{\beta+n}]. \quad (19)$$

Their weight functions are

$$w(x) = (1-x)^\alpha (1+x)^\beta. \quad (20)$$

The domain of orthogonality of this class of polynomials is a finite interval, for which, without loss of generality, one can take a basic interval $(-1, 1)$ [10].

3 Idea and Motivation

Special functions find their application in solving numerous problems in mathematics, physics and technical sciences. Therefore, there is a need for an efficient yet straightforward way to calculate the values of these functions. This need leads to the concept of a modern, intuitive application that delivers strong performance and enables users to approximate the value of a function with minimal steps. This concept has been previously explored. For example, articles like [5] analyse similar software used until the end of the twentieth century, discussing their characteristics, problems, and prospects for further development.

Several popular applications that offer a vast range of calculations also include special functions, but they often present them in a slightly different format than intended for this application – typically involving more steps, a less intuitive interface, etc. Additionally, existing solutions often demand extra effort to use, such as learning notation, syntax, or even knowledge of basic programming concepts.